

DAV UNIVERSITY JALANDHAR
SYNOPSIS OF
“A RESPONSIVE COFFEE WEBSITE”
Submitted to the
FACULTY OF SCIENCE



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INTRODUCTION

The “Responsive Coffee Website” is a high-performance web application designed to bridge the gap between ethical coffee sourcing and modern digital commerce. Built with a focus on transparency and sustainability, the platform allows users to explore specialty coffees while learning about the farmers and ecosystems behind each bean. The project follows a complete development lifecycle, moving from a high-fidelity Figma prototype to a functional HTML/CSS/JS frontend, backed by a robust MySQL database.

System Analysis

Identification and Formulation of the problem:

Traditional coffee retail often lacks transparency regarding supply chains, leading to a "disconnected" consumer experience. Furthermore, many e-commerce sites are cluttered with "dark patterns" that encourage impulsive, non-responsible purchasing. There is a specific need for a digital platform that allows customers to:

- Visualize the ethical journey of the coffee product from farm to cup.
- Provide a clean, distraction-free interface to reduce "digital waste" and choice paralysis.
- Maintain accurate, secure records of sustainable inventory via a structured MySQL database.
- Track order status in real-time with transparent communication.

Need and Significance of Proposed Work:

In the traditional coffee market, consumers are often disconnected from the origin of their beans, and record-keeping is often manual and prone to error. This project is significant because:

- **Digital Transparency:** It transforms the routine act of buying coffee into a mindful, transparent experience.
- **Data Integrity:** Using a computerized MySQL system ensures that every "Responsible" claim is backed by real-time inventory and supply chain data.
- **Market Reach:** It enables ethical farmers and small-scale roasters to expand their services to a national audience, increasing potential revenue.

Objectives

The primary goal is to design and develop a fully functional, responsive, and user-friendly web application for ethical coffee trade:

DETAILED OBJECTIVE:

UI/UX Design and Prototyping (Figma Phase)

To design a **Minimalist-Ethical interface** in Figma that avoids "dark patterns," ensuring a stress-free and transparent user journey.

To create a **Mobile-First Responsive Design**, ensuring that the ethical coffee catalog is accessible and aesthetically consistent across all devices (Smartphones, Tablets, and Desktops).

To develop **Interactive Wireframes** that map out the "Traceability Flow," allowing users to click on a coffee bag and see its journey from the farm.

Frontend Engineering (Development Phase)

To implement a Dynamic Product Catalog using HTML5, CSS3, and JavaScript that allows users to filter coffee by roast level, origin, and "Responsibility Score."

To build an Asynchronous Shopping Cart (using AJAX/JavaScript) that updates prices and quantities in real-time without requiring a page refresh, enhancing user retention.

To integrate Data Visualization Components (like progress bars or maps) that show the environmental impact or fair-trade status of each purchase.

Backend and Database Management (MySQL Phase)

To architect a Relational Database Schema in MySQL that ensures data integrity for thousands of products, user profiles, and transaction records.

To implement Secure User Authentication, ensuring that customer data and purchase histories are protected using modern encryption standards.

Scheduling and Methodology of Proposed Work

Development Methodology:

The “Responsible Coffee Website” follows the Agile Software Development Methodology, an iterative and incremental approach. Instead of a linear path, the project is broken down into small, manageable "Sprints." This is particularly effective for your project because:

Reasons for Choosing Agile Methodology:

- It allows flexibility in incorporating changes in requirements.
 - It ensures continuous testing and quality improvement.
 - It promotes better collaboration between developers and stakeholders.
 - It reduces development risks by delivering working components in phases.
 - It improves overall project transparency and efficiency.
 - Integrate secure payment gateway.
 - Ensure data encryption and security measures.
1. **Flexibility:** It allows you to refine the "Responsible" branding and ethical sourcing features based on visual feedback from the Figma stage.
 2. **Continuous Integration:** By developing the MySQL backend and Frontend in parallel Sprints, data connectivity can be tested early and often.
 3. **Risk Reduction:** Working components (like a functional login or product grid) are delivered in phases, ensuring the core e-commerce engine works before adding complex features.

Detailed Project Schedule (24 Weeks):

Weeks 1-4: The Design & Logic Foundation

Sprint 1 (Requirements): Finalizing the "Responsible Coffee" feature list, such as carbon footprint tracking and farm-to-cup transparency.

Sprint 2 (UI/UX): Developing high-fidelity prototypes in Figma to establish the minimalist aesthetic.

Weeks 5-12: Frontend Construction

Sprint 3 (Static Layouts): Converting Figma designs into responsive HTML5 and CSS3 structures.

Sprint 4 (Interactivity): Using JavaScript to handle the dynamic shopping cart and real-time sustainability score updates.

Weeks 13-20: Backend Architecture & Data Persistence

Sprint 5 (Database Design): Setting up the MySQL environment and creating normalized tables for Users, Ethical Products, and Orders.

Sprint 6 (API Integration): Connecting the frontend to the backend logic to allow for secure user login and data retrieval.

Weeks 13-20: Backend Architecture & Data Persistence

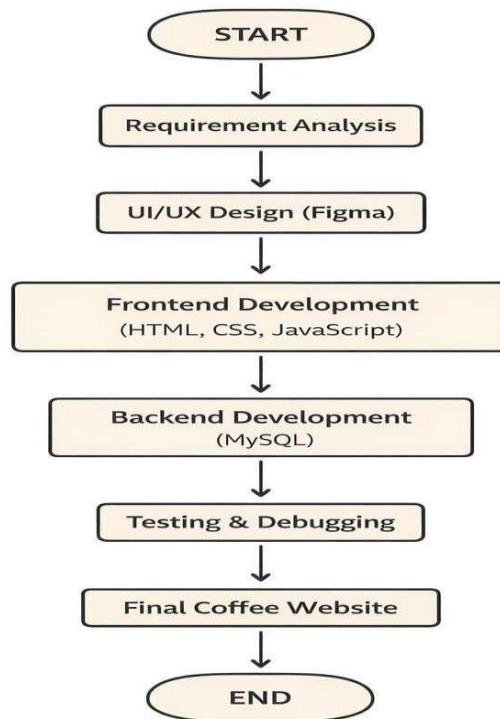
Sprint 5 (Database Design): Setting up the MySQL environment and creating normalized tables for Users, Ethical Products, and Orders.

Sprint 6 (API Integration): Connecting the frontend to the backend logic to allow for secure user login and data retrieval.

Hardware and Software Requirements

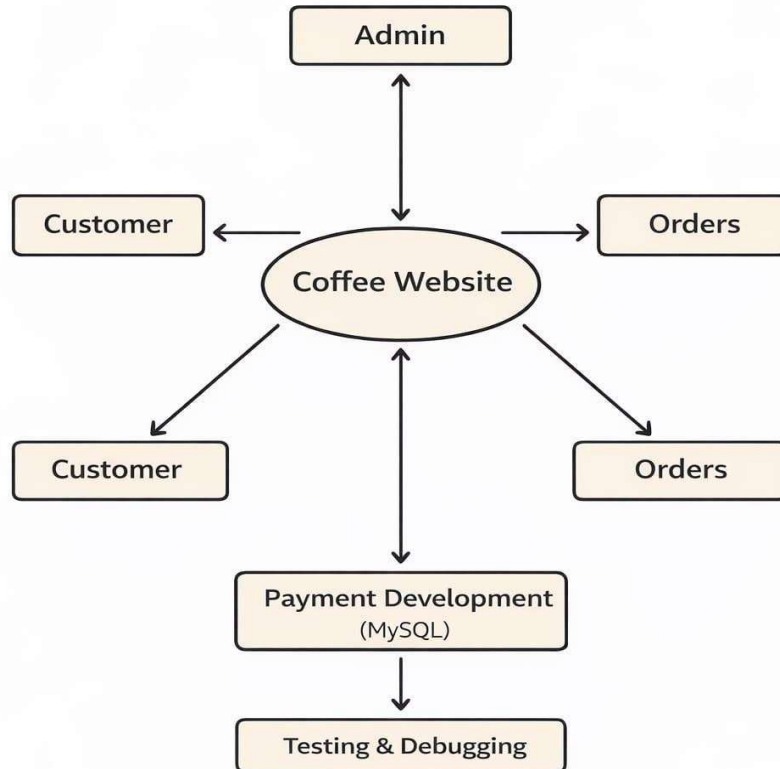
- **Hardware Requirements:**
 - **Processor:** AMD RYZEN 7.
 - **RAM:** Minimum 4 GB.
 - **Storage:** 500 GB or above.
- **Software Requirements:**
 - **Design:** Figma.
 - **Frontend:** HTML5, CSS3, JavaScript.
 - **Database:** MySQL (Relational Database Management System).
 - **Development Tools:** VS Code and Google Chrome

Diagram Flowchart Diagram- Coffee Website project



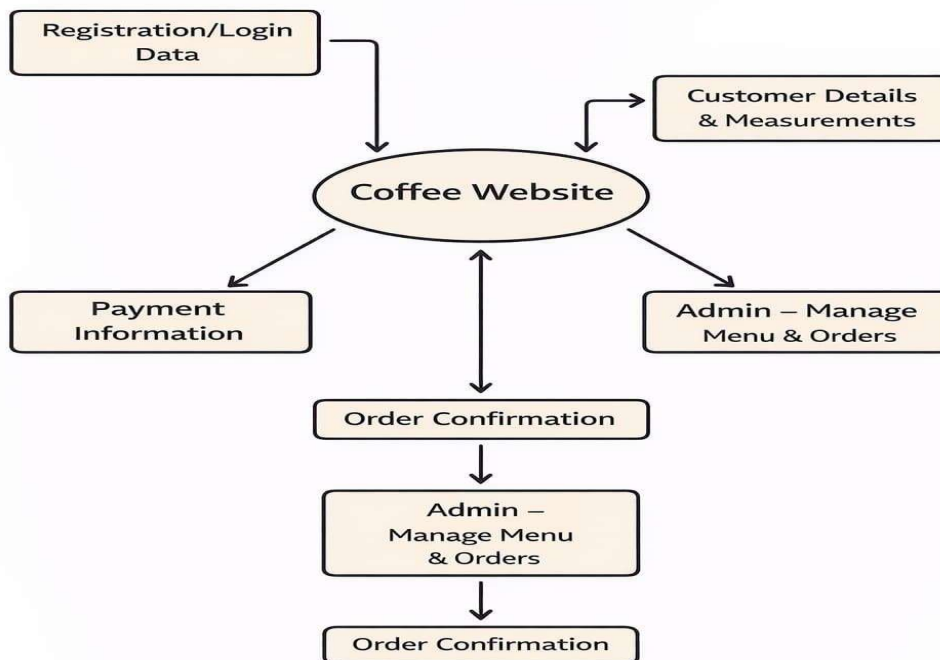
Zero Level DFD (Context Diagram)

Zero Level DFD | Context Diagram



Level 1 DFD (Detailed Diagram)

Level 1 DFD (Detailed DFD)



SYSTEM DESIGN AND ARCHITECTURE

Module Details User Module: Handles registration, secure login, and ethical loyalty point management.

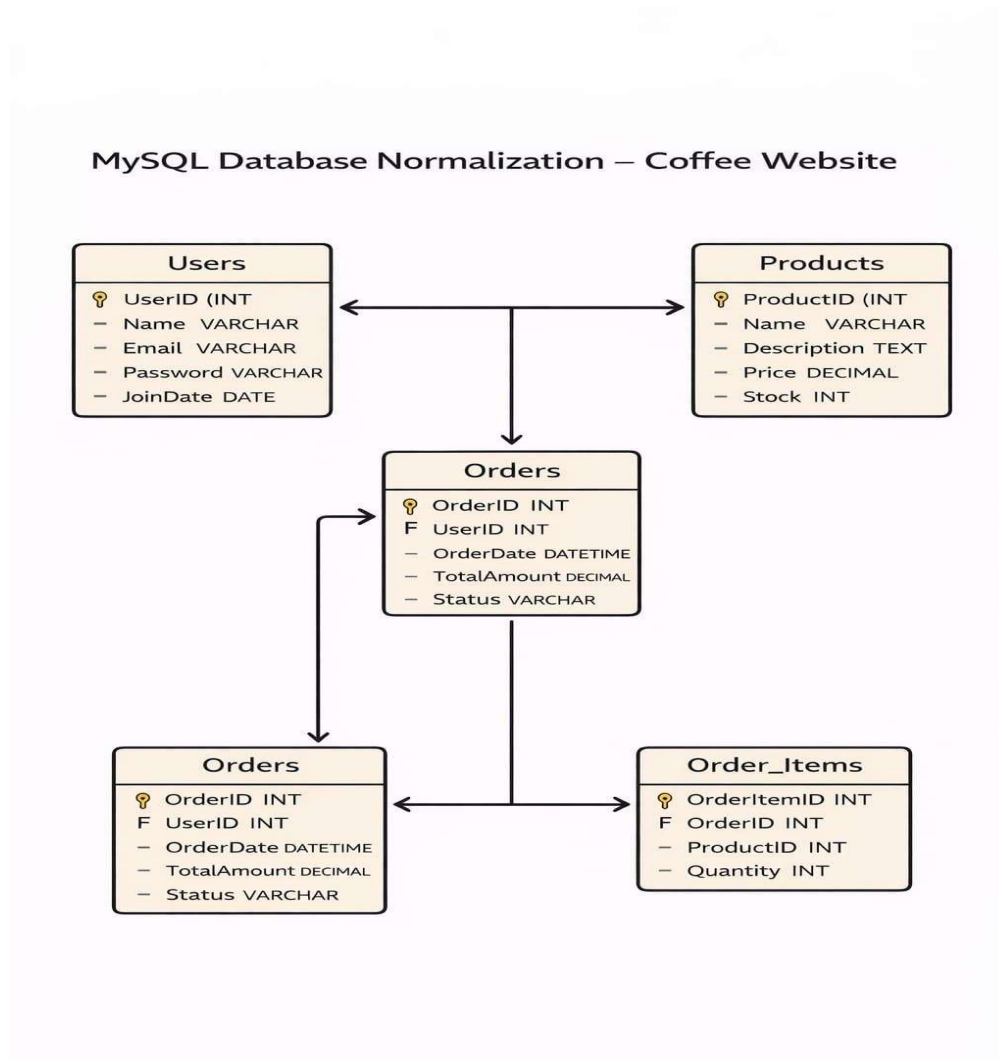
Product & Ethics Module: Manages coffee roast profiles, origin data, and sustainability scores.

Order & Payment Module: Facilitates secure transactions and real-time order tracking.

Admin Module: Allows administrators to manage stock levels and update farm traceability details.

Normalized MySQL Database Schema

To ensure academic depth, the database is normalized to the 3rd Normal Form (3NF):



Conclusion

The development of the “Responsible Coffee Website” represents a significant step forward in the digital transformation of the specialty coffee industry. By successfully migrating from a traditional, manual retail model to a sophisticated full-stack web application, this project demonstrates how technology can be leveraged to solve real-world problems related to supply chain transparency and consumer trust.